

Simplifying Surds

AQA · KS4 Year 10–11

Answer each question in your workbook. Show your working.

VISUAL EXAMPLE

$$\begin{aligned}\sqrt{72} &= \sqrt{(36 \times 2)} \\ &= \sqrt{36} \times \sqrt{2} \\ &= 6\sqrt{2}\end{aligned}$$

Simplifying a surd: split off the largest square factor.

COMMON MISTAKES

- ✗ Not using the **LARGEST** square factor — do it in one step, not several.
- ✗ $\sqrt{a + b} \neq \sqrt{a} + \sqrt{b}$. The rule only works for multiplication.
- ✗ $3\sqrt{12} = 3 \times 2\sqrt{3} = 6\sqrt{3}$, not $3\sqrt{12}$.
- ✗ $\sqrt{a} \times \sqrt{a} = a$, not $\sqrt{(a^2)}$ left unsimplified.

METHOD

- 1 Find the largest square factor of the number
- 2 Split: $\sqrt{n} = \sqrt{(\text{square} \times \text{rest})}$
- 3 Use $\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$
- 4 Square-root the perfect square out
- 5 Answer: $k\sqrt{m}$ (m has no square factors)

WORKED EXAMPLE

Simplify $\sqrt{72}$.

Step 1 — Largest square factor: 36

Step 2 — $\sqrt{72} = \sqrt{(36 \times 2)} = \sqrt{36} \times \sqrt{2} = 6\sqrt{2}$

VISUAL AID

1. (2 marks)

Simplify: a) $\sqrt{12}$ b) $\sqrt{18}$ c) $\sqrt{20}$ d) $\sqrt{50}$

2. (2 marks)

Simplify: a) $\sqrt{8}$ b) $\sqrt{32}$ c) $\sqrt{45}$ d) $\sqrt{75}$

3. (3 marks)

Simplify: a) $\sqrt{98}$ b) $\sqrt{200}$ c) $\sqrt{128}$ d) $\sqrt{300}$

4. (3 marks)

Simplify: a) $\sqrt{4 \times 7}$ b) $3\sqrt{12}$ c) $2\sqrt{50}$ d) $5\sqrt{8}$

5. (3 marks)

Simplify fully: a) $\sqrt{48} + \sqrt{12}$ b) $\sqrt{75} - \sqrt{27}$ c) $2\sqrt{18} + 3\sqrt{2}$

6. (4 marks)

Show: a) $\sqrt{80} = 4\sqrt{5}$ b) $3\sqrt{28} = 6\sqrt{7}$ c) $\sqrt{180} = 6\sqrt{5}$

7. (4 marks)

Square area = 72 cm^2 . a) Exact side? b) Perimeter? c) Compare with square of side $3\sqrt{2}$.

8. (5 marks)

a) Show $\sqrt{a^2b} = a\sqrt{b}$. b) Simplify $\sqrt{50x^4y^2}$. c) Right triangle sides $\sqrt{18}$, $\sqrt{32}$ — exact hypotenuse?
